

Generative AI on AWS

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Amazon Web Services

Agenda

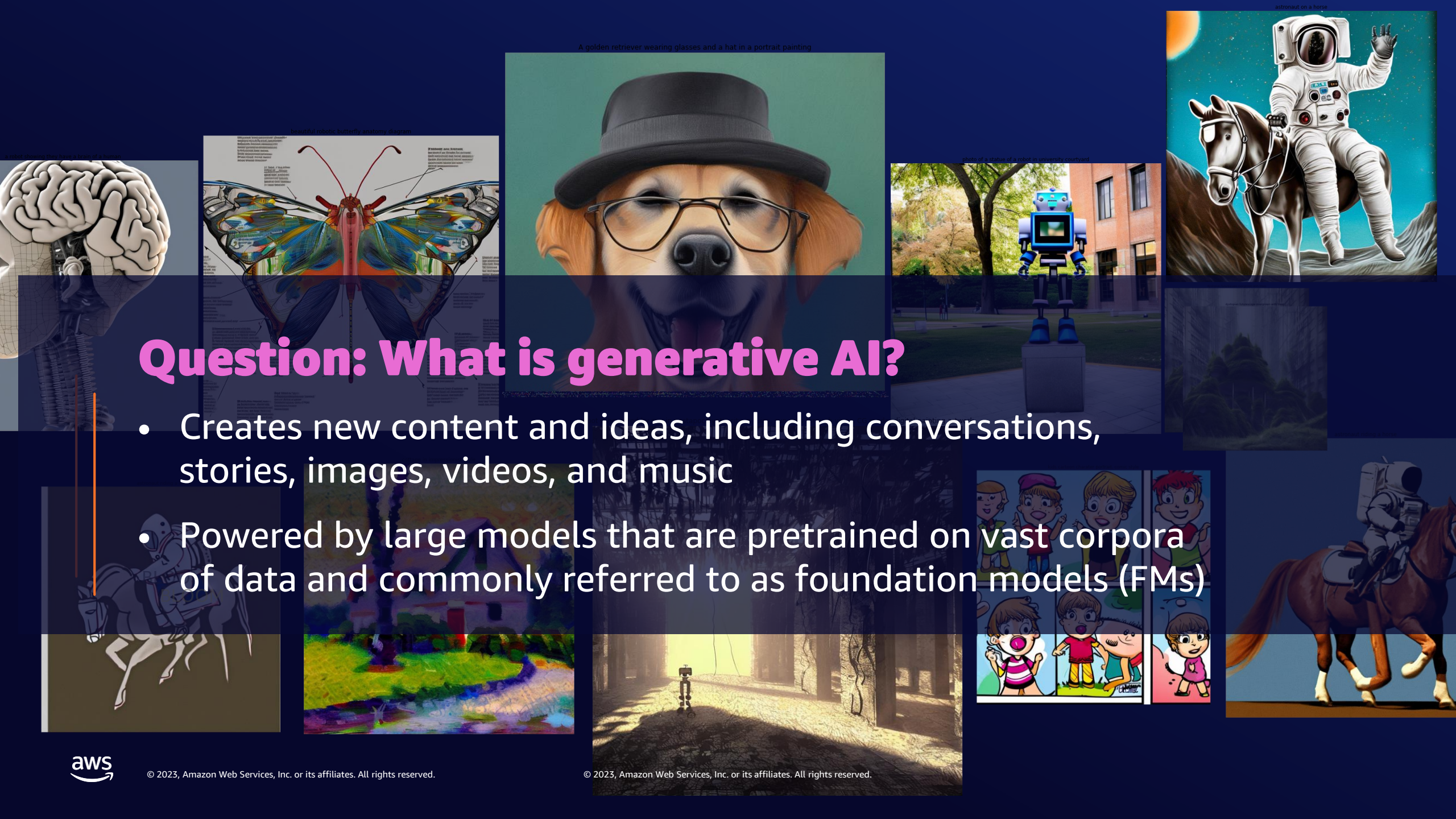
- Introduction to Generative AI
- ReInventing with Generative AI on AWS
- Amazon Bedrock AI Stylist Demo & Architecture Walkthrough
- Build a GenAI based code generator using Amazon Bedrock



Innovation can
transform industries



GENERATIVE AI



A golden retriever wearing glasses and a hat in a portrait painting

beautiful robot: butterfly anatomy diagram

photo of a statue of a robot in university courtyard

astronaut on a horse

Question: What is generative AI?

- Creates new content and ideas, including conversations, stories, images, videos, and music
- Powered by large models that are pretrained on vast corpora of data and commonly referred to as foundation models (FMs)

AI

MACHINE
LEARNING

SIMPLE

INPUTS



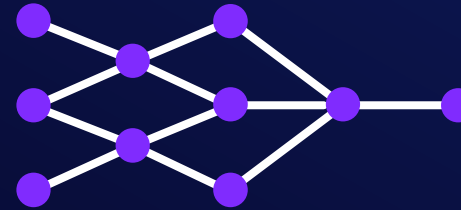
SIMPLE

OUTPUTS

DEEP
LEARNING

COMPLEX

INPUTS



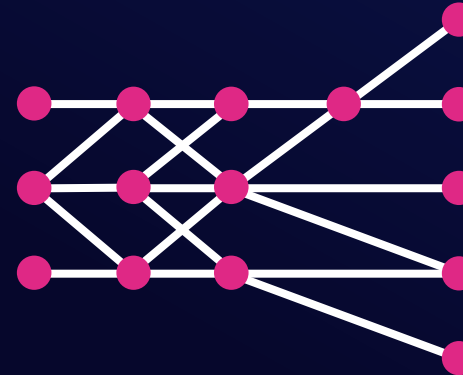
SIMPLE

OUTPUTS

FOUNDATION
MODELS

COMPLEX

INPUTS

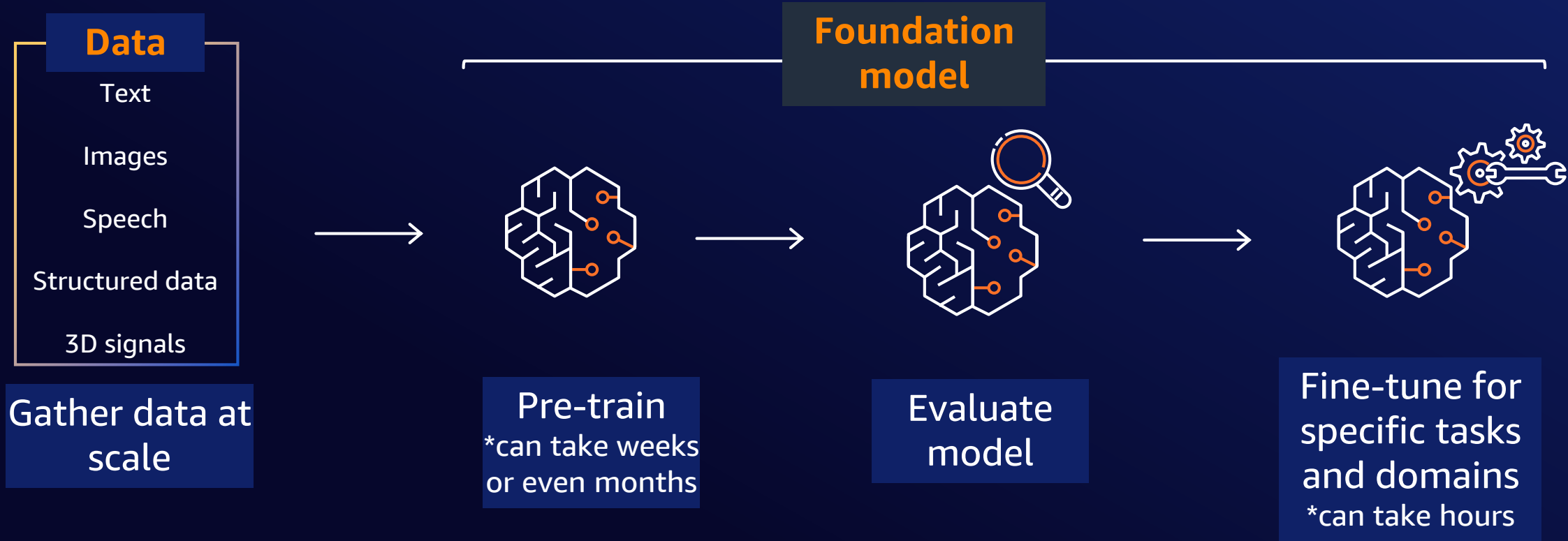


COMPLEX

OUTPUTS



How does a foundation model work?



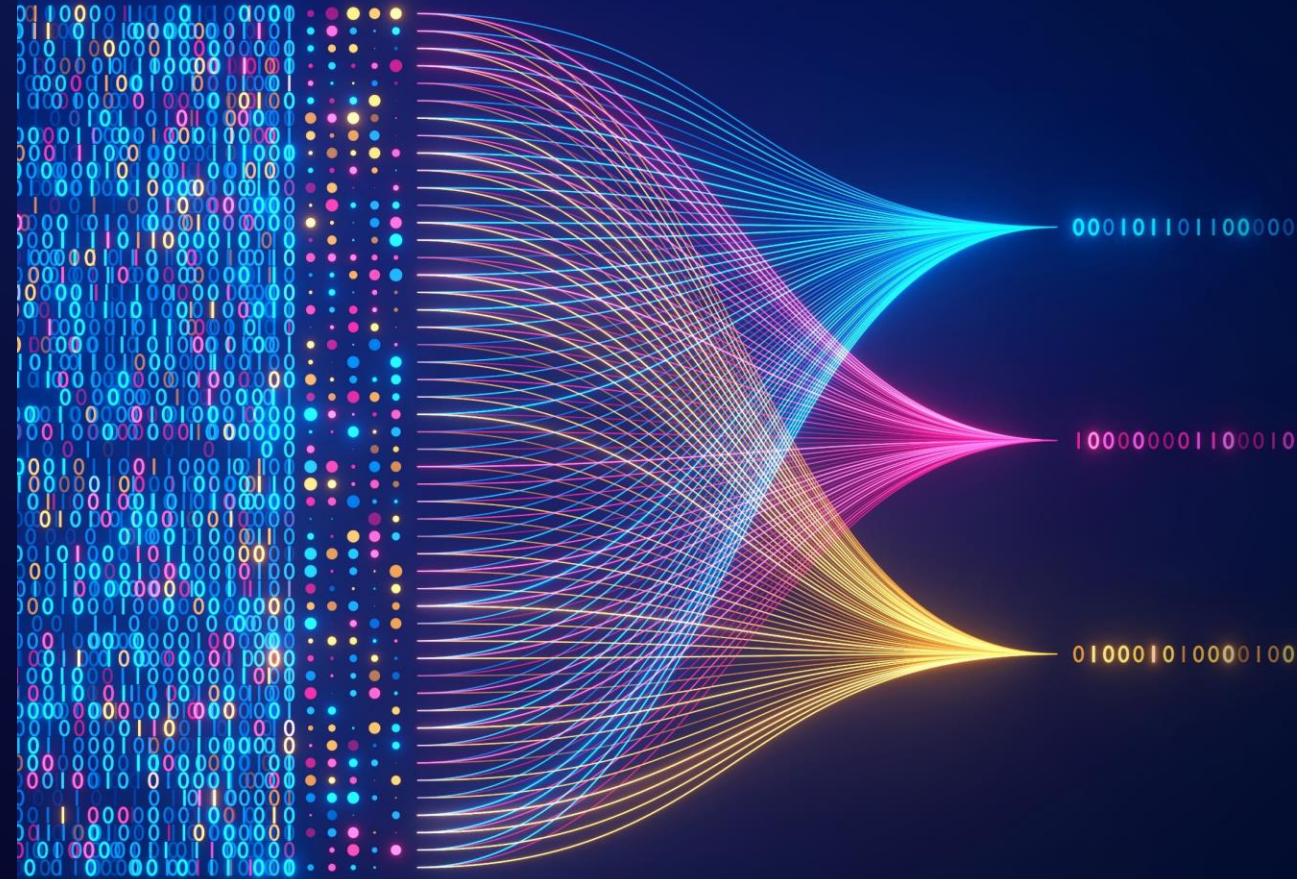
Generative AI is powered by foundation models

Pretrained on vast amounts of unstructured data

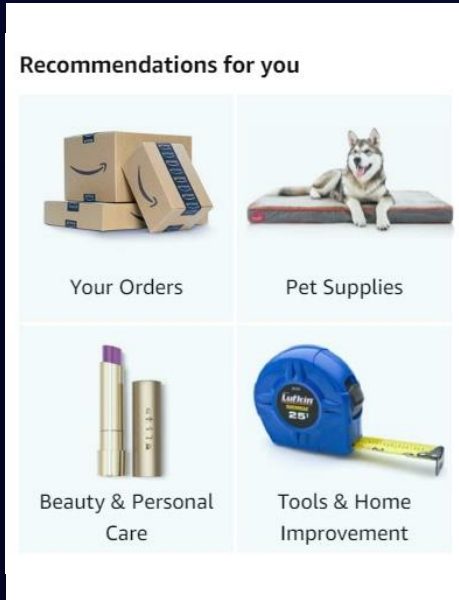
Contain large number of parameters that make them capable of learning complex concepts

Can be applied in a wide range of contexts

Customize FMs using your data for domain specific tasks



ML innovation is in Amazon's DNA



**4,000 products
per minute** sold
on Amazon.com



1.6M packages
every day



Billions of Alexa
interactions each week



Just Walk Out
technology in airports,
stadiums and more



Reinventing with generative AI

Generative AI Stack



APPLICATIONS THAT LEVERAGE LLMs AND OTHER FMs

TOOLS TO BUILD WITH LLMs AND OTHER FMs

INFRASTRUCTURE FOR FM TRAINING AND INFERENCE



Generative AI Stack



APPLICATIONS THAT LEVERAGE LLMs AND OTHER
FMs

TOOLS TO BUILD WITH LLMs AND OTHER FMs

INFRASTRUCTURE FOR FM TRAINING AND INFERENCE





CG1

NVIDIA Tesla
M2050 "Fermi"
GPUs

G2

NVIDIA GRID
GK104 "Kepler"
GPUs

P2

NVIDIA
K80
GPUs

G3

NVIDIA
Tesla M60
GPUs

P3

NVIDIA V100
Tensor Core
GPUs

G4

NVIDIA T4
Tensor Core
GPUs

P4

NVIDIA A100
Tensor Core
GPUs

G5

NVIDIA A10G
Tensor Core
GPUs

G5g

NVIDIA T4G
Tensor Core
GPUs

P5

NVIDIA H100
Tensor Core
GPUs

Innovating at the silicon level

AWS Trainium



AWS Inferentia

4x

HIGHER THROUGHPUT



10x

LOWER LATENCY

Generative AI Stack

APPLICATIONS THAT LEVERAGE LLMs AND OTHER FMs

TOOLS TO BUILD WITH LLMs AND OTHER FMs

INFRASTRUCTURE FOR FM TRAINING AND INFERENCE



GPUs



Trainium



Inferentia



SageMaker



UltraClusters



EFA



EC2 Capacity Blocks



Nitro



Neuron



Amazon Bedrock

The easiest way to build and scale generative AI applications with LLMs and other FMs

Choice of industry-leading FMs from AI21 Labs, Amazon, Anthropic, Cohere, Meta, and Stability AI

Customize FMs using your organization's data

Enterprise-grade security and privacy

Amazon Bedrock

Broad choice of models

AI21labs

amazon

ANTHROPIC

cohere

Meta

stability.ai

JURASSIC-2

AMAZON TITAN

CLAUDE

COMMAND + EMBED

LLAMA 2

STABLE DIFFUSION XL

Generative AI Stack

APPLICATIONS THAT LEVERAGE LLMs AND OTHER FMs

TOOLS TO BUILD WITH LLMs AND OTHER FMs



Amazon Bedrock

Guardrails | Agents | Customization Capabilities

INFRASTRUCTURE FOR FM TRAINING AND INFERENCE



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 **Amazon Bedrock**

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NEW

Amazon Q

A generative AI-powered assistant for work that is tailored to your business

AVAILABLE IN PREVIEW

Provides interactive answers, solves problems, generates content, and takes action

Understands your company information, code, and systems

Personalizes interactions based on your role and permissions

Built to be secure and private

Amazon Q is



AMAZON Q
YOUR AWS EXPERT



AMAZON Q
YOUR BUSINESS EXPERT



AMAZON Q IN AMAZON QUICKSIGHT
YOUR BI EXPERT



AMAZON Q IN AMAZON CONNECT
YOUR CONTACT CENTER EXPERT

Generative AI Stack

APPLICATIONS THAT LEVERAGE LLMs AND OTHER FMs



Amazon Q



Amazon Q in
Amazon QuickSight



Amazon Q in
Amazon Connect



Amazon
CodeWhisperer

TOOLS TO BUILD WITH LLMs AND OTHER FMs



Amazon Bedrock

Guardrails

Agents

Customization Capabilities

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GPUs



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Nitro



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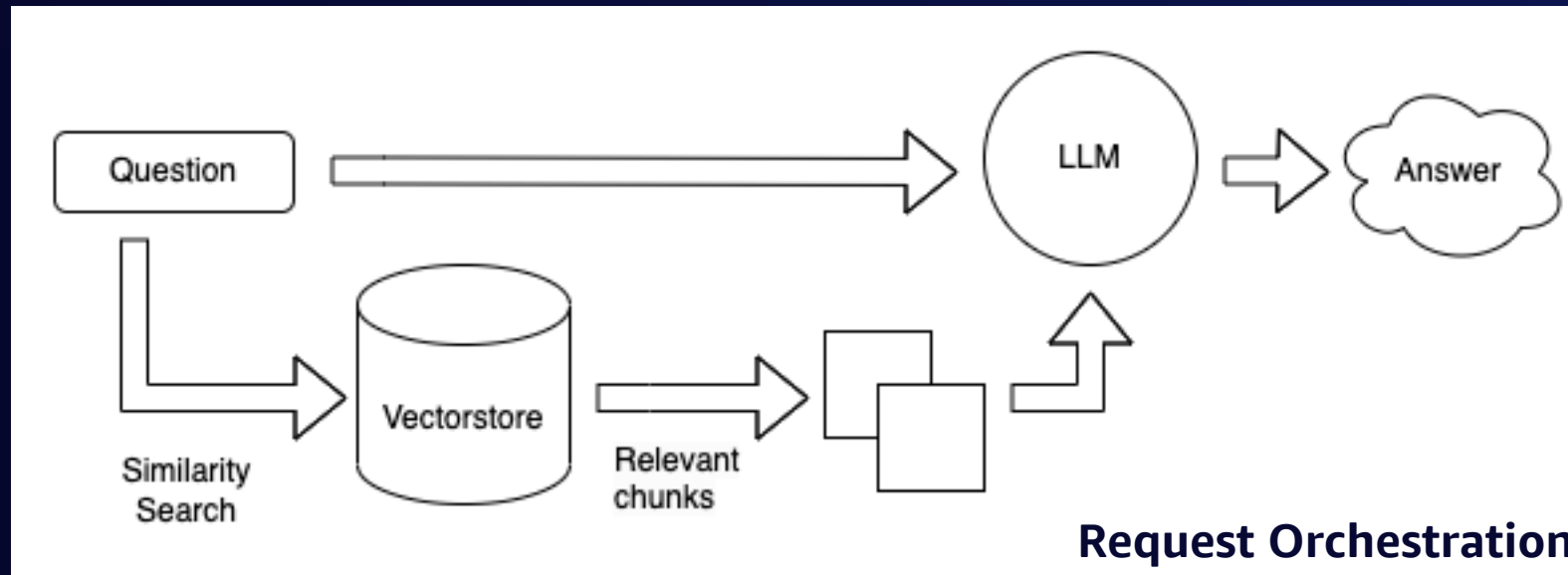
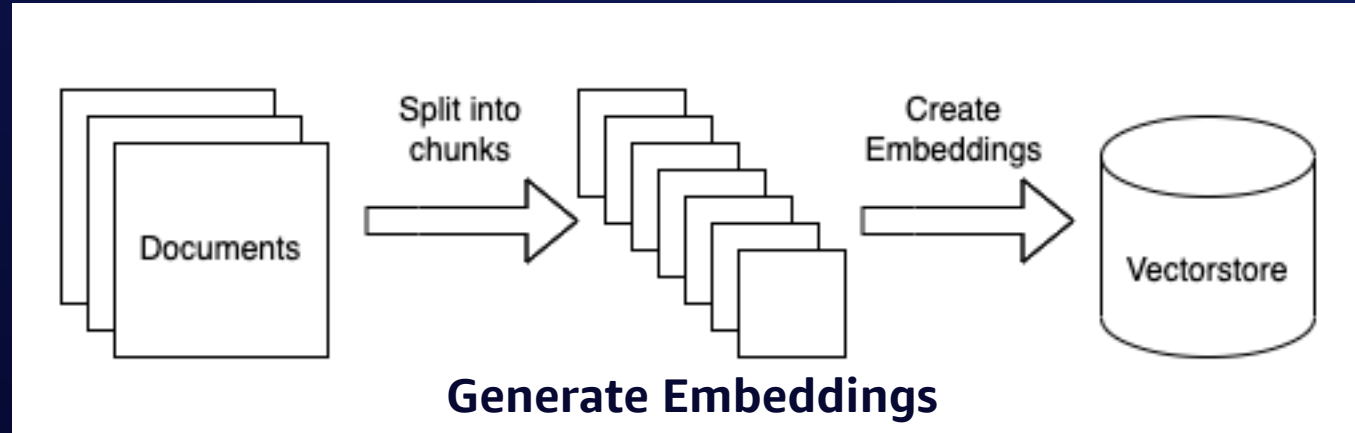
See it in Action

AIStylist Demo

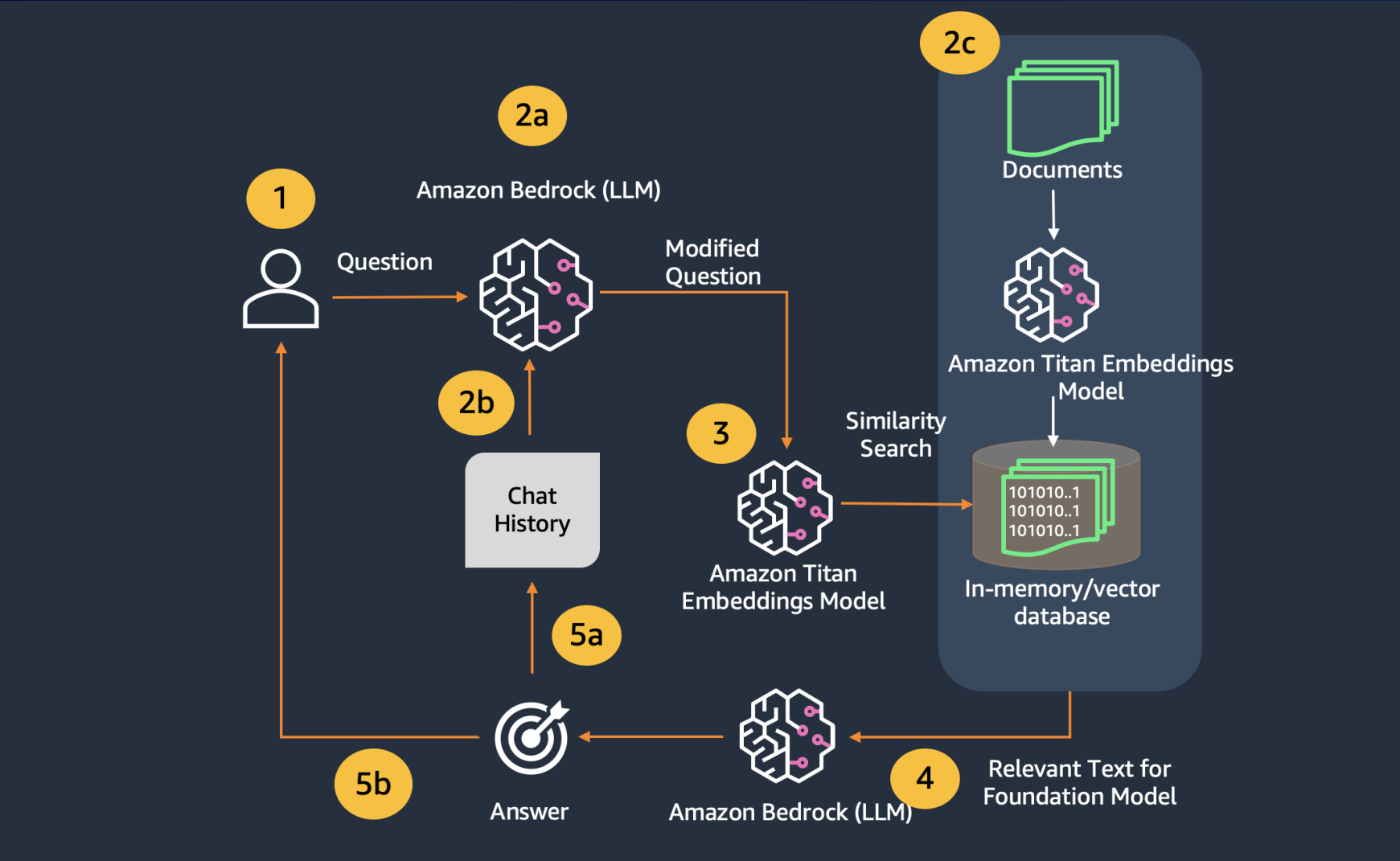
<https://aistylist.awsplayer.com/>



AiStylist Solution Components (RAG & Embeddings)



AiStylist Solution Architecture



Knowledge bases/APIs ⓘ

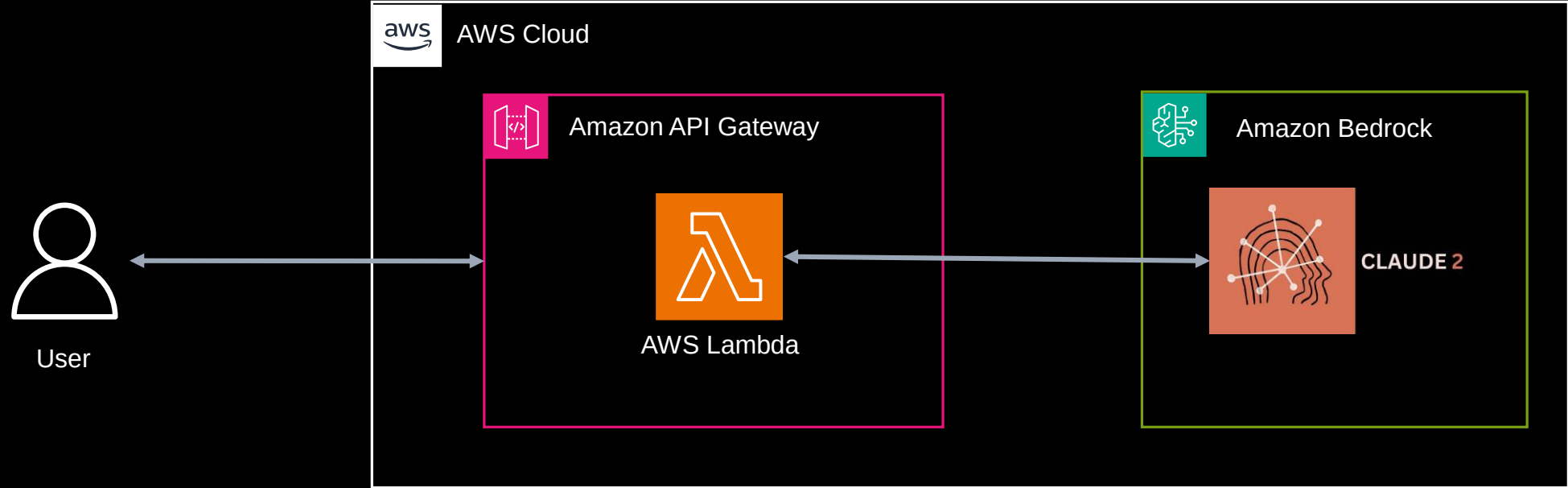
Product catalog	private data
Fashion trends	public data
Order history	private data
Customer reviews	private data

Add more knowledge bases



Lets Build a Code Generator

Code Generator



Learning Resources

Amazon Bedrock Workshop



Hands-on Course on Generative AI with Large Language Models



AWS Learning Needs Analysis:

Build a data-driven plan to accelerate learning



Learn more about AWS Skill Builder:



Additional Reading

[REACT: Synergizing Reasoning and Acting in Language Models](#)

[Chain-of-Thought Prompting Elicits Reasoning in Large Language Models](#)

[Attention is All You Need](#)



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